

ADP AI - Prediction Report

Generated by ADP AI Random Forest Prediction Pipeline

Report ID: 120e63d2-0357-48f0-b5f3-e6856fb8e19d

Algorithm: Random Forest Regressor

Model Type: regression

Accuracy Score: 96.73%

Dataset Details

Target Column: Sales

Total Rows: 500

Total Columns: 8

Model Selection

AI Detected Intent: auto

Final Task Type: regression

Decision Reason: The target column is numeric and continuous, so ADP AI selected Random Forest Regressor.

Metrics

R2 Score: 0.9673

Mean Absolute Error: 11382.3176

Root Mean Squared Error: 14096.5629

Training Rows: 400

Testing Rows: 100

Top Feature Importance

Profit: 0.83016

UnitPrice: 0.084986

UnitsSold: 0.072632

MarketingSpend: 0.004585

Date: 0.004033

Product: 0.001915

Region: 0.001689

Prediction Samples

Actual: 88756.0 | Predicted: 97290.6533

Actual: 83047.0 | Predicted: 59233.4867

Actual: 49129.0 | Predicted: 80386.7667

Actual: 112970.0 | Predicted: 107175.2267

Actual: 46178.0 | Predicted: 49386.59

Actual: 24774.0 | Predicted: 21486.4567

Actual: 128579.0 | Predicted: 116289.62

Actual: 128596.0 | Predicted: 131923.37

Actual: 92422.0 | Predicted: 109324.5767

Actual: 39257.0 | Predicted: 56802.2867

Smart Recommendations

1. Data quality: No major missing values found in the uploaded dataset.
2. No duplicate rows found, so the dataset is suitable for prediction analysis.

3. Target analysis: Average Sales is 135256.25, median is 112716.5, minimum is 2875.0, and maximum is 419104.0.
4. High priority: Improve 'Profit' because it has a positive relationship with Sales. Random Forest importance is 83.02%.
5. High priority: Improve 'UnitPrice' because it has a positive relationship with Sales. Random Forest importance is 8.5%.
6. High priority: Improve 'UnitsSold' because it has a positive relationship with Sales. Random Forest importance is 7.26%.
7. High priority: Improve 'MarketingSpend' because it has a positive relationship with Sales. Random Forest importance is 0.46%.
8. Best performing group in 'Date' is '2026-05-25' with average Sales of 419104.0. Follow patterns from this group.
9. Sales improvement: Focus on customer segments, products, or regions with higher revenue contribution.
10. Growth strategy: Increase campaigns for high-performing products and improve weak-performing categories.

Prediction Flow

1. User uploaded dataset
2. ADP AI read rows and columns
3. ADP AI checked target column
4. ADP AI cleaned missing values
5. ADP AI prepared numeric and categorical features
6. ADP AI selected suitable ML model
7. Random Forest model trained using train-test split
8. Model predictions generated
9. Evaluation metrics calculated
10. Feature importance generated
11. Recommendation intelligence generated
12. Prediction report prepared